

ACOUSTIC SCENE CLASSIFICATION USING SPATIAL FEATURES

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- ▶ 4th-order Ambisonic acoustic scene recordings
- ▶ Spectral and spatial analysis
- ▶ Machine listening system
- ▶ Spatial features outperform spectral

OUTLINE





Soundman OKM II Klassik Electret
<http://www.soundman.de/en>

- ▶ Use of mono/stereo recordings
- ▶ Easier to record (maybe)
- ▶ Inheritance of work from other fields
- ▶ Applications in smart devices / robotics
- ▶ Limited to spectral / binaural spatial features
- ▶ Can work well (83.3 % accuracy in latest DCASE)

PREVIOUS WORK

Aircraft takeoff: 110 dB(A)



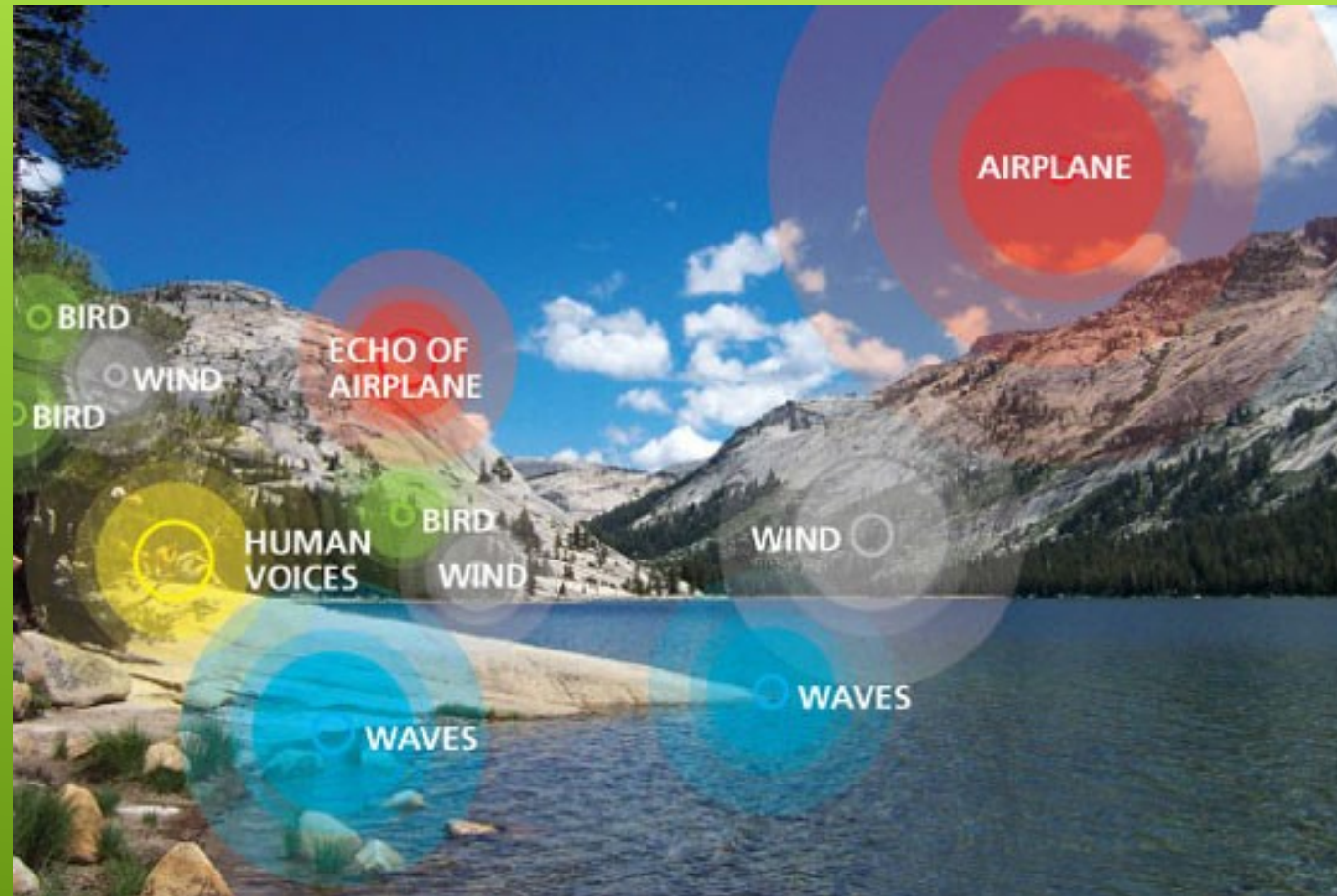
<http://www.flydayton.com/>

Rock concert: 110 dB(A)



<http://ajr.org/2014/11/13/music-critics-role-changing/>

ALTERNATIVE APPLICATION – ENVIRONMENTAL SOUND

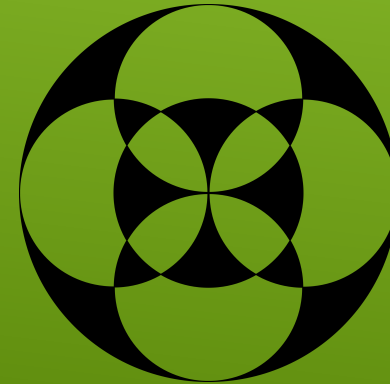
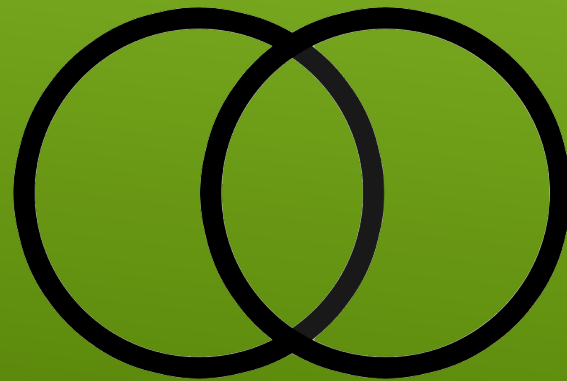


<https://www.nps.gov/yose/learn/nature/soundscape.htm>

- ▶ Goal: event detection
- ▶ Scene classification as first step

- ▶ Lots of examples of each scene needed
- ▶ TUT database – stereo
- ▶ DEMAND database – not enough examples
- ▶ **New database** required

DATABASE





- ▶ mh Acoustics Eigenmike
- ▶ 4th-order Ambisonics – high spatial resolution
- ▶ 8 examples of 8 different scene classes
- ▶ 10 minutes per clip
- ▶ 24-bit / 48 kHz
- ▶ Available now!

DOI [10.5281/zenodo.1012809](https://doi.org/10.5281/zenodo.1012809)

Beach
Busy Street
Park
Pedestrian Zone
Quiet Street
Shopping Centre
Train Station
Woodland

EIGENSCAPE DATABASE



EigenScape Recording Map

Map detailing proposed locations for recording of the EigenScape dataset.

[more](#)

163 views

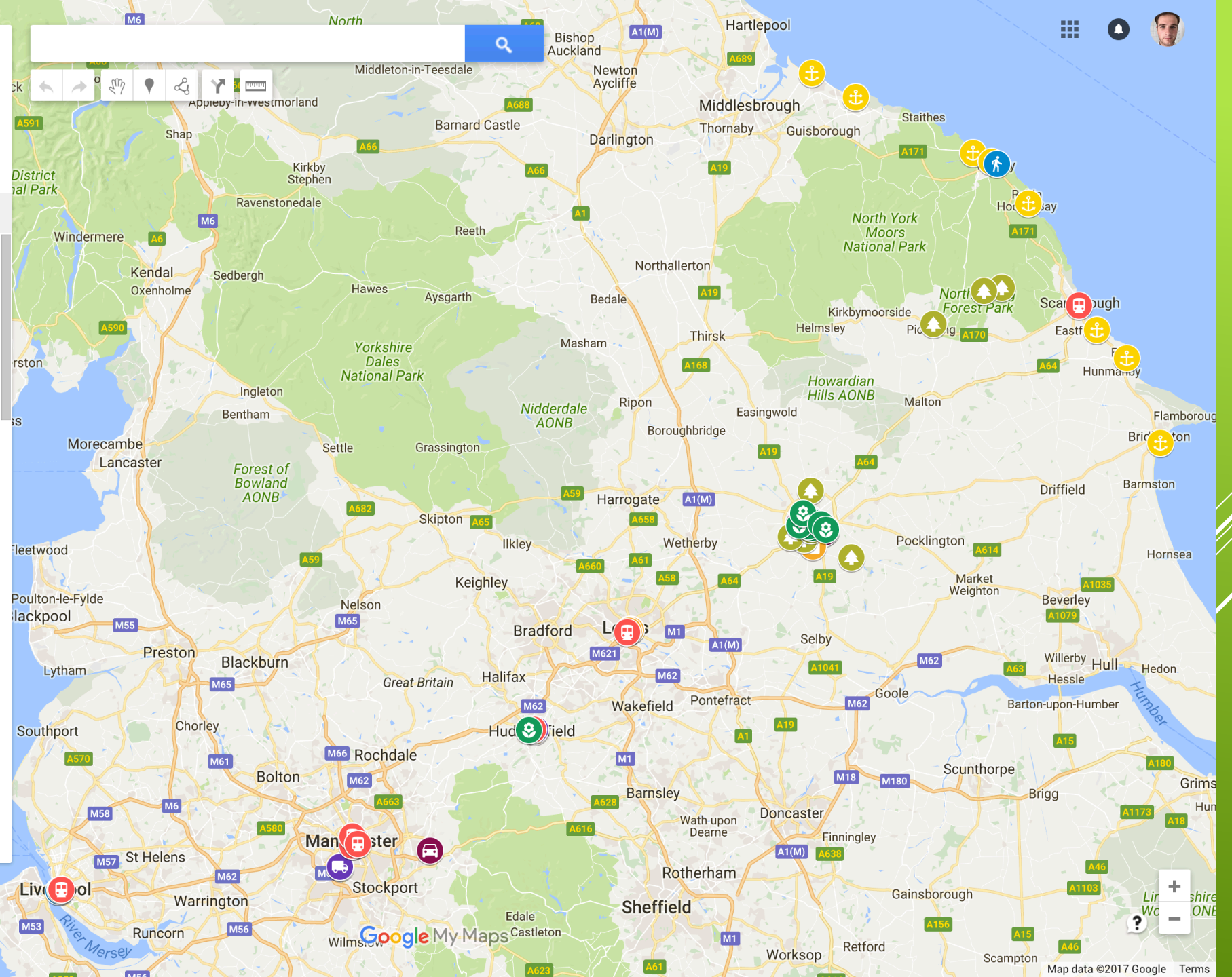
All changes saved in Drive

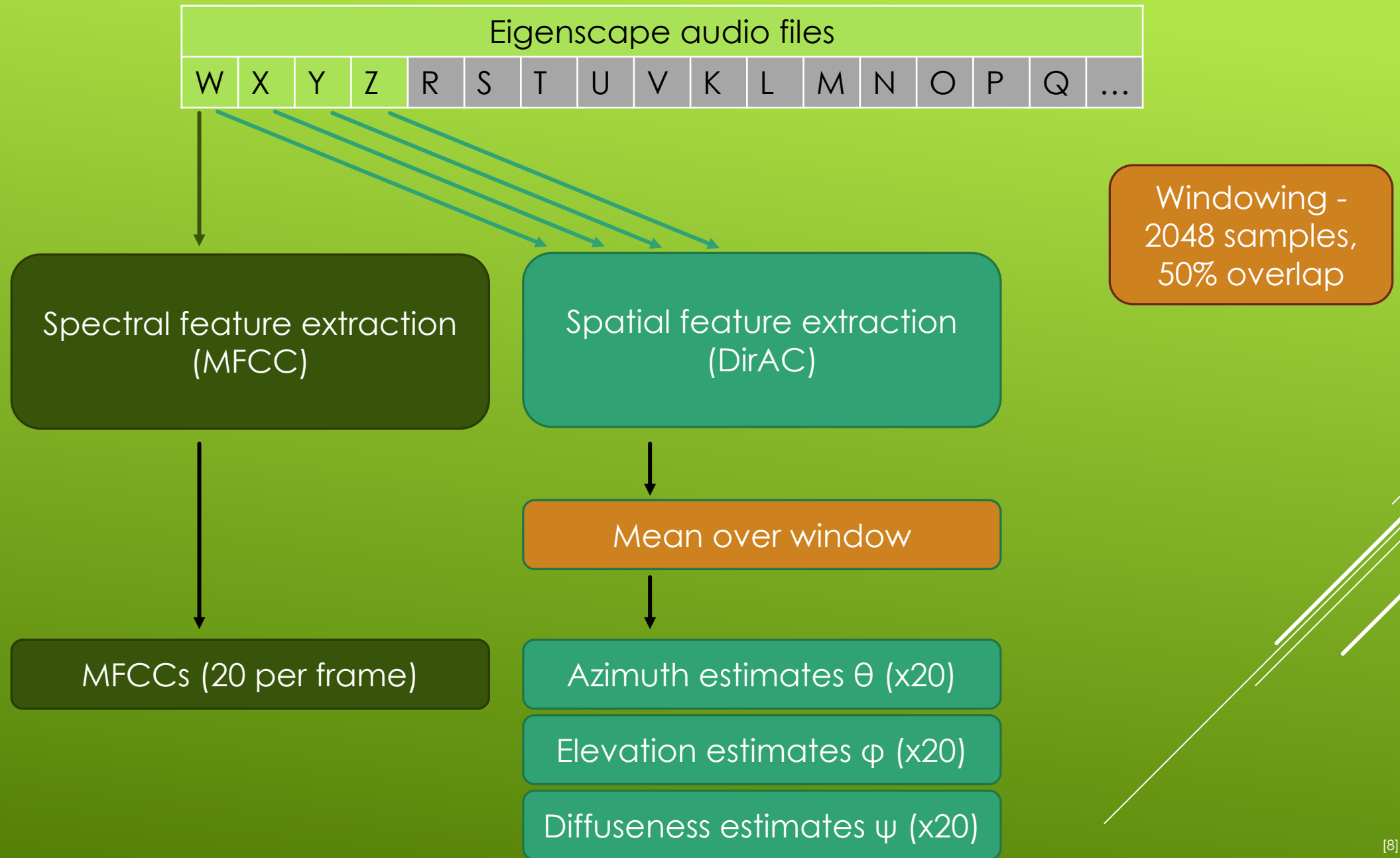
 Add layer  Share  Preview

☒ Locations

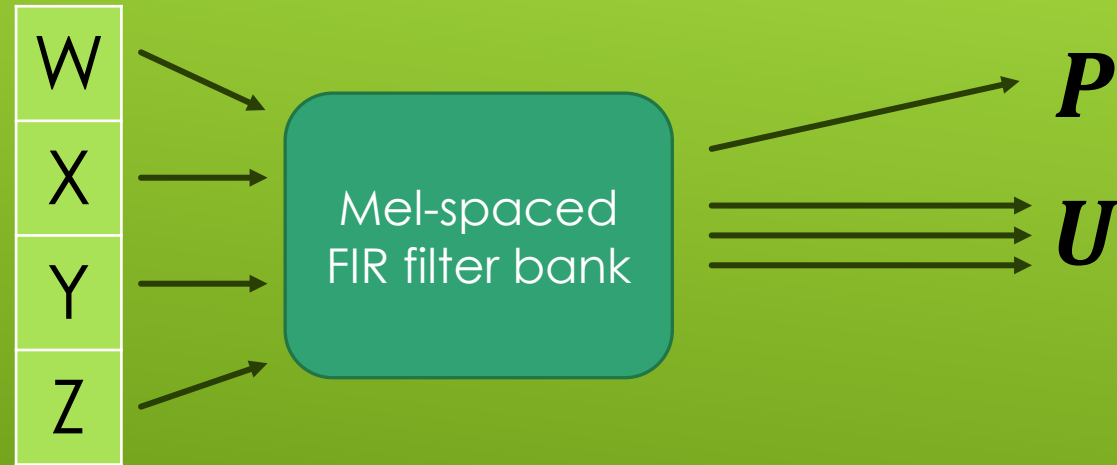
 Individual styles

-  Bridlington Beach
-  Filey Beach
-  Cayton Bay
-  Redcar Beach
-  Saltburn Beach
-  Sandsend
-  Whitby West Cliff
-  Robin Hood's Bay
-  Chatfield Road
-  Thomas Street
-  Rising Moon, Matley Lane
-  Church Lane
-  Main Street
-  St. Benedict Road
-  Windmill Rise Corner
-  Holmefield Lane
-  Clayton Square
-  Church Street
-  Shambles Square
-  Market Street
-  Church Street, Whitby





DIRECTIONAL AUDIO CODING (DIRAC)



$$\mathbf{D} = -\mathbf{P}\mathbf{U}$$

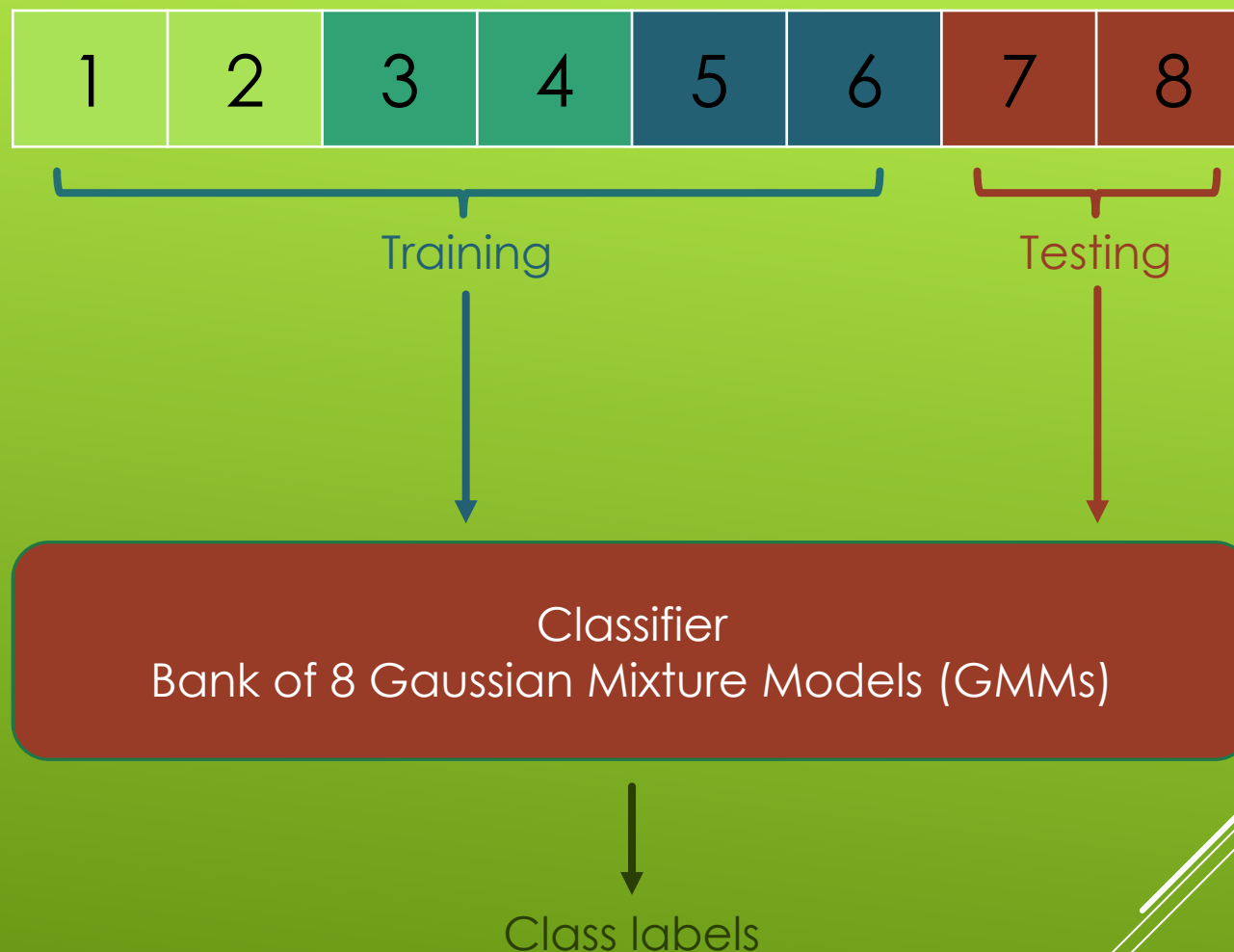
$\nearrow$ Azimuth
 $\rightarrow$ Elevation

$$\varphi = 1 - \frac{\|\mathbf{D}\|}{c\{\mathbf{E}\}}$$

Diffuseness

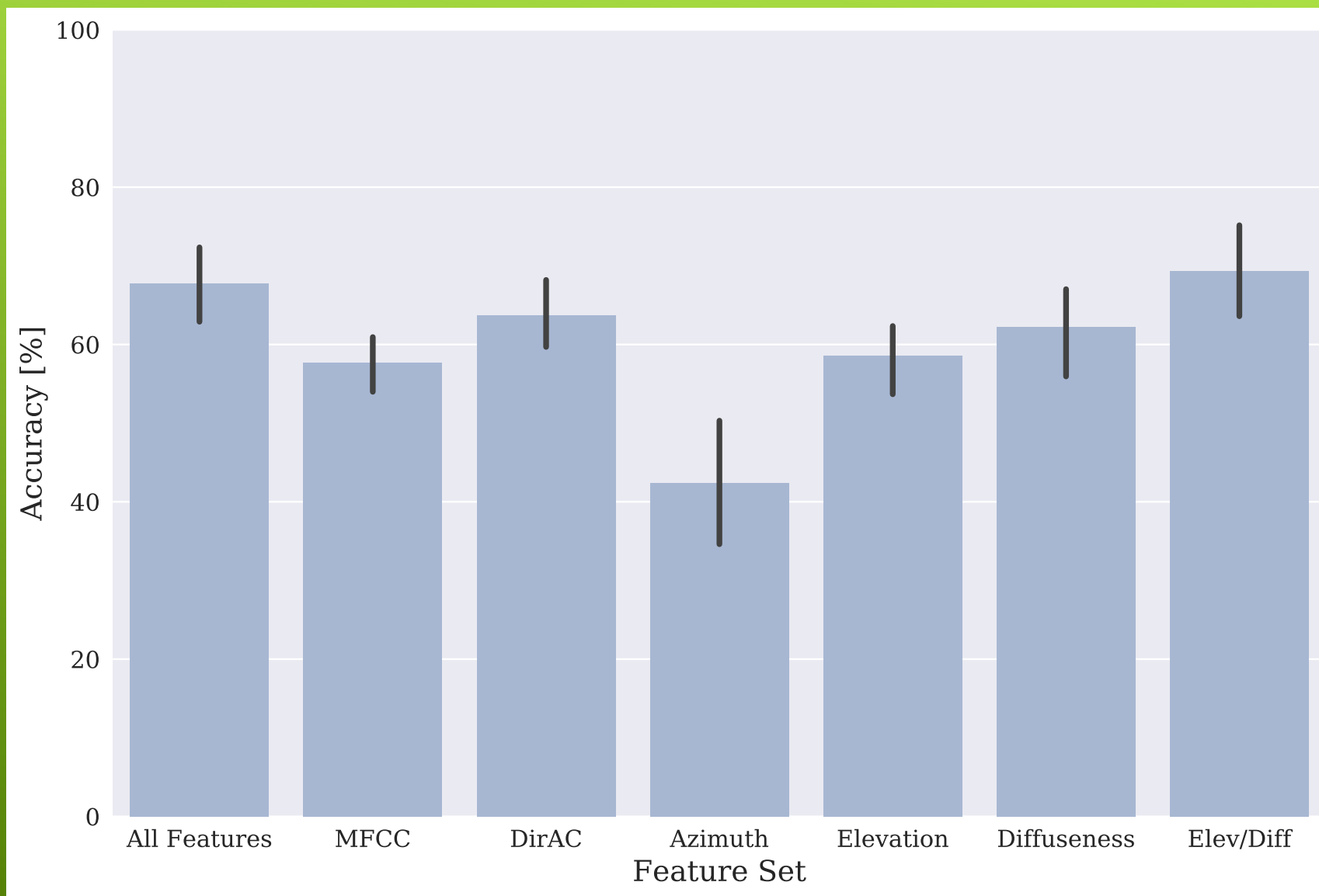
$$\mathbf{E} = \frac{1}{2}p_0 \left(\frac{\mathbf{P}^2}{Z_0^2} + \|\mathbf{U}\|^2 \right)$$

4-fold cross-validation e.g.

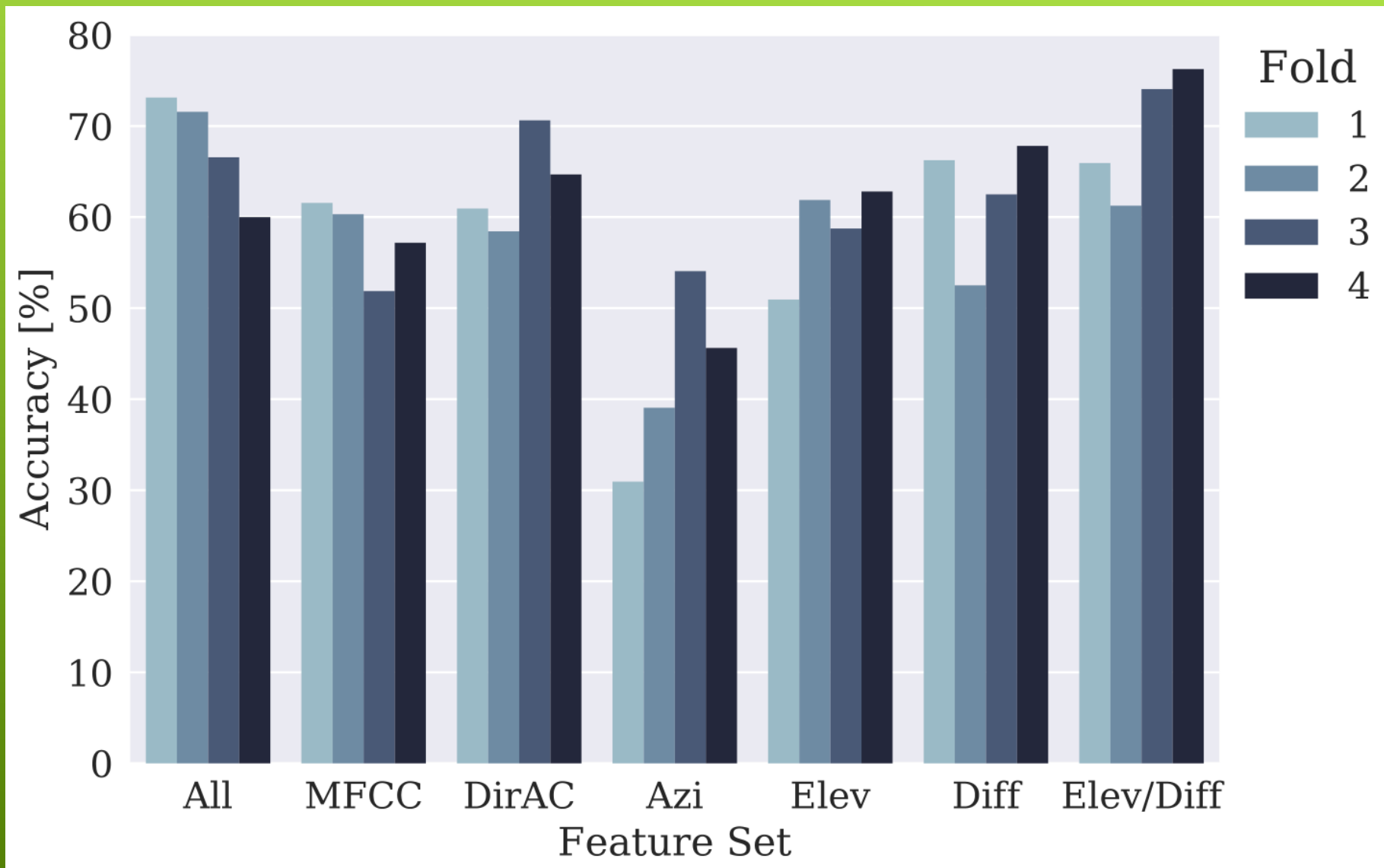


- Probabilities summed across 30 seconds of segments
- Highest probability returned determines label selected
- Classifiers trained and tested across all four folds with results aggregated

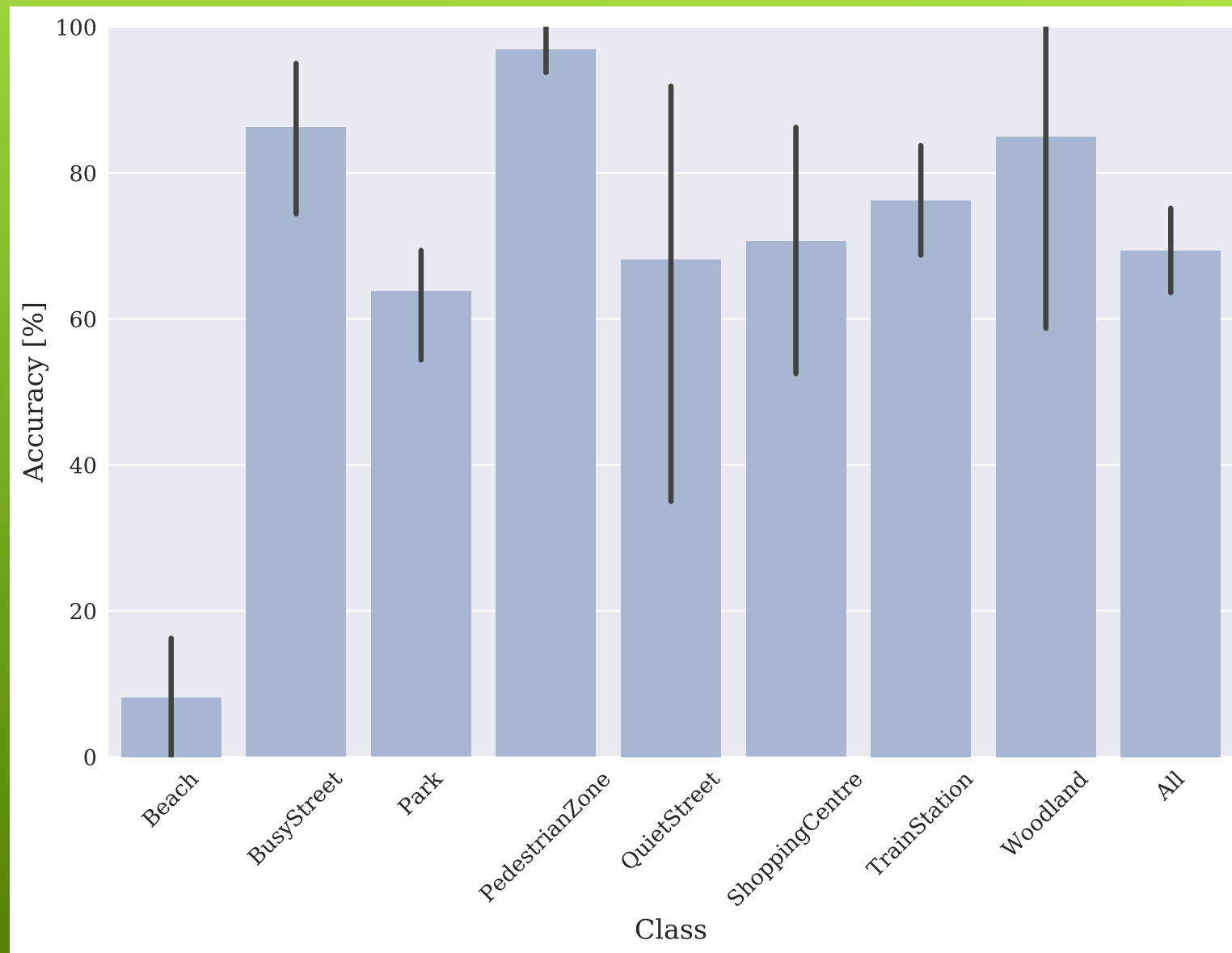
Mean classification accuracies using MFCC and DirAC features



Classification accuracies using MFCC and DirAC features

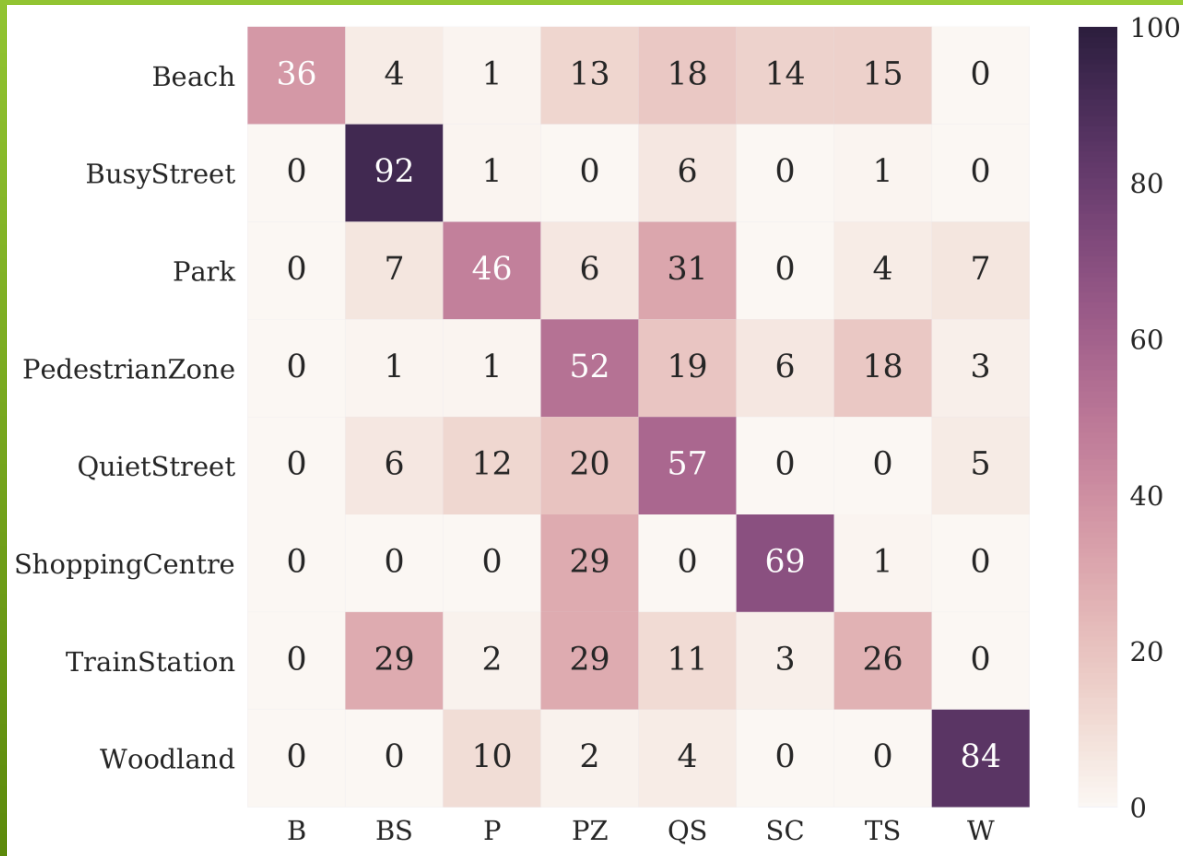


Per-class accuracies using Elevation/Diffuseness features

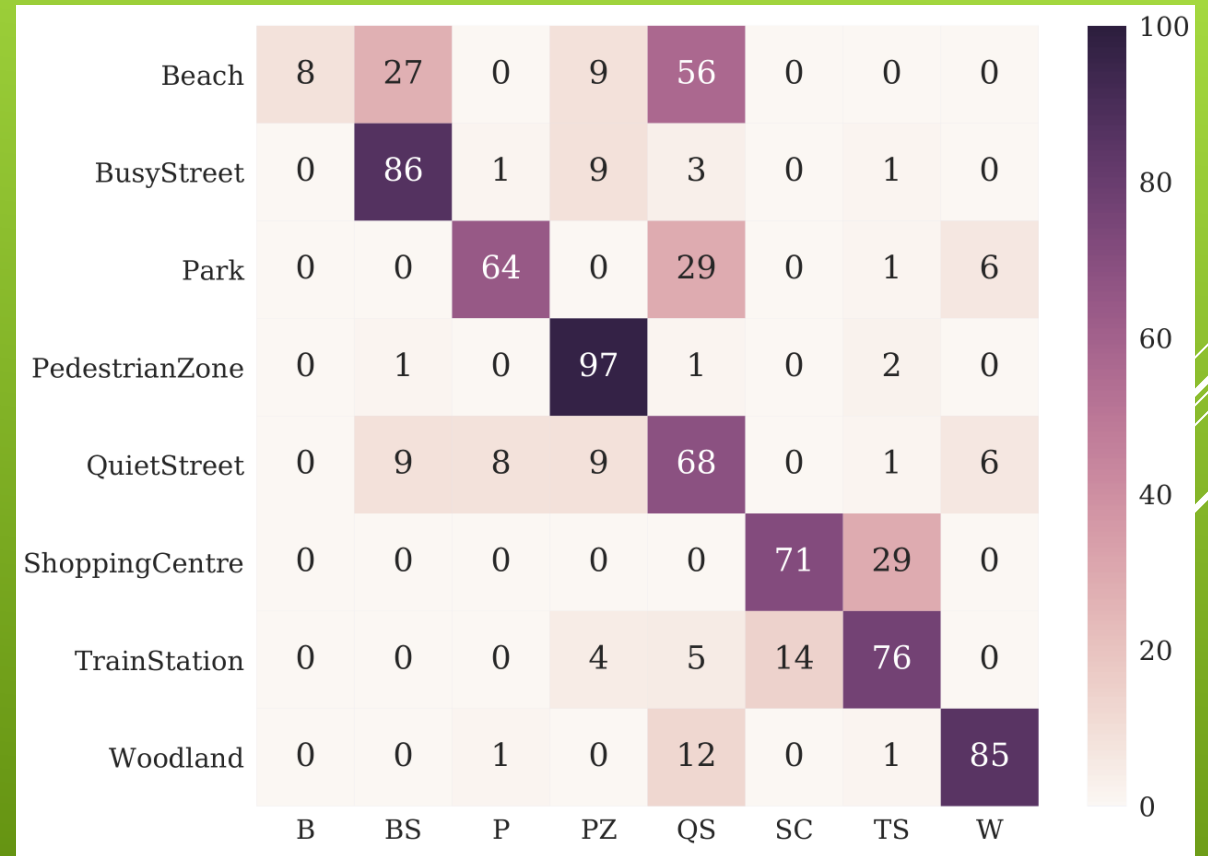


Confusion Matrices

MFCC features



Elevation/Diffuseness features



- ▶ Accurate classification with spatial features
- ▶ E/D features outperform MFCC
- ▶ Important initial result – spatial features valuable
- ▶ *Spectral* similarity and *spatial* similarity not the same
- ▶ Good, not perfect accuracy validates EigenScape

- ▶ **Next steps:**
 - ▶ Use of higher-order channels
 - ▶ Event detection

CONCLUSIONS

- [1] mh Acoustics, *em32 Eigenmike microphone array release notes*, mh acoustics, 25 Summit Ave, Summit, NJ 07901, April 2013. [Online]. Available: <https://mhacoustics.com/sites/default/files/EigenmikeReleaseNotesV15.pdf>
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- [10] Marc Ciufu Green, & Damian Murphy. (2017). EigenScape [Data set]. Zenodo. <http://doi.org/10.5281/zenodo.1012809>

REFERENCES

- ▶ EigenScape Database: <http://doi.org/10.5281/zenodo.1012809>
- ▶ Code: <https://github.com/marc1701/EigenScape>
- ▶ Map: <http://bit.ly/EigenSMap>

LINKS